

Talk Notes on Coates–Sujatha, Section 3.4–3.6

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Main Goal of the Topic

The goal of Sections 3.4–3.6 is to translate the exact sequence of Coleman power series from Chapter 2 into an exact sequence of p -adic measures on

$$G = \text{Gal}(\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p).$$

The central result is Theorem 3.5.1, the fundamental exact sequence

$$0 \longrightarrow \mu_{p-1} \times T_p(\mu) \longrightarrow U_\infty \xrightarrow{\tilde{L}} \Lambda(G) \xrightarrow{\beta} T_p(\mu) \longrightarrow 0. \quad (3.11)$$

This builds a dictionary between three kinds of objects:

$$\begin{aligned} &\text{norm-compatible local units } u \in U_\infty, \\ &\text{Coleman power series } f_u(T) \in R^\times, \\ &p\text{-adic measures on } G. \end{aligned}$$

Section 3 explains which measures on the additive group $\mathbb{Z}_p$ are actually supported on the units $\mathbb{Z}_p^\times$: under the Mahler transform this is the simple condition $\psi = 0$. This is necessary because Mahler transform naturally lives on $\mathbb{Z}_p$, while the Galois group G is identified with $\mathbb{Z}_p^\times$.

Section 4 then transports the exact sequence for Coleman power series to a measure-theoretic exact sequence. The map $\tilde{L}$ takes a unit system u and produces a measure on G . Integrating powers of the cyclotomic character against this measure recovers the logarithmic derivatives $\delta_k(u)$ up to the elementary factor $1 - p^{k-1}$:

$$\int_G \chi(g)^k d\tilde{L}(u) = (1 - p^{k-1})\delta_k(u). \quad (3.12)$$

Section 6 uses the decomposition

$$\Lambda(G) \simeq \bigoplus_{i \bmod p-1} R$$

to turn the image question for δ_k into evaluation of power series at $T = \chi(\gamma)^k - 1$.

Basic Definitions to Keep in Mind

Let

$$K_\infty = \mathbb{Q}_p(\mu_{p^\infty}) = \bigcup_{n \geq 0} \mathbb{Q}_p(\mu_{p^{n+1}}), \quad G = \text{Gal}(K_\infty/\mathbb{Q}_p).$$

For $\sigma \in G$ and every p -power root of unity ζ , there is a unique element $\chi(\sigma) \in \mathbb{Z}_p^\times$ such that

$$\sigma(\zeta) = \zeta^{\chi(\sigma)}.$$

The resulting homomorphism $\chi : G \longrightarrow \mathbb{Z}_p^\times$ is called the cyclotomic character. It is an isomorphism.

The notation $T_p(\mu)$ means the Tate module of p -power roots of unity:

$$T_p(\mu) = \varprojlim_n \mu_{p^{n+1}},$$

where the transition maps are $x \mapsto x^p$. If we choose a compatible system $(\zeta_n)_n$ with $\zeta_{n+1}^p = \zeta_n$, then every $a \in \mathbb{Z}_p$ defines an element

$$(\zeta_n^a)_n \in T_p(\mu).$$

This is the identification $\mathbb{Z}_p \simeq T_p(\mu)$ used in the final map β of the exact sequence.

Finally, when $a \in \mathbb{Z}_p$, the expression $(1+T)^a$ is defined by the binomial series

$$(1+T)^a = \sum_{n \geq 0} \binom{a}{n} T^n \in \mathbb{Z}_p[[T]].$$

This makes sense because $\binom{a}{n} \in \mathbb{Z}_p$ for all $a \in \mathbb{Z}_p$.

1 Notation and Objects

Notation	Meaning
p	A fixed odd prime.
K_∞	The local cyclotomic extension $\mathbb{Q}_p(\mu_{p^\infty})$.
G	$\text{Gal}(K_\infty/\mathbb{Q}_p)$. The cyclotomic character gives $G \simeq \mathbb{Z}_p^\times$.
χ	Cyclotomic character, defined by $\sigma(\zeta) = \zeta^{\chi(\sigma)}$ for all p -power roots of unity ζ .
$T_p(\mu)$	The inverse limit of p -power roots of unity under $x \mapsto x^p$. After choosing (ζ_n) , it is identified with $\mathbb{Z}_p$ by $a \mapsto (\zeta_n^a)_n$.
R	The power series ring $\mathbb{Z}_p[[T]]$.
D	The differential operator $D = (1+T) \frac{d}{dT}$.
φ	The endomorphism of R given by $\varphi(f)(T) = f((1+T)^p - 1)$.
ψ	The trace-type operator of Proposition 2.2.3.
N	The norm operator of Proposition 2.2.3.
W	The group $\{f \in R^\times : N(f) = f\}$.
U_∞	Projective limit of local units in the cyclotomic tower, with norm maps.
$f_u(T)$	Coleman power series attached to $u = (u_n) \in U_\infty$, uniquely determined by $f_u(\zeta_n - 1) = u_n$.
$\Delta(f)$	Logarithmic derivative $\Delta(f) = (1+T)f'(T)/f(T)$.
δ_k	Higher logarithmic derivative: $\delta_k(u) = \left(D^{k-1}\Delta(f_u)\right)_{T=0}$.
$\Lambda(H)$	Iwasawa algebra of a profinite abelian group H .
M	Mahler transform $M : \Lambda(\mathbb{Z}_p) \rightarrow R$.
Υ	The inverse Mahler transform $R \rightarrow \Lambda(\mathbb{Z}_p)$.
$\tilde{\chi}$	Isomorphism $\Lambda(G) \simeq \Lambda(\mathbb{Z}_p^\times)$ induced by $\chi : G \simeq \mathbb{Z}_p^\times$.
$\widetilde{M}$	The isomorphism $\Lambda(G) \rightarrow R^{\psi=0}$ defined by $\widetilde{M} = M \circ \tilde{\chi}$.
$\tilde{L}$	The measure-valued map $\tilde{L}(u) = \widetilde{M}^{-1}(L(f_u))$.
ϑ	The decomposition map $\Lambda(G) \simeq \bigoplus_{i \bmod p-1} R$ used in Section 3.6.

2 Earlier Results One Should Use

Here is the dependency list for the talk.

1. **Theorem 2.1.2.** For every $u = (u_n) \in U_\infty$, there is a unique $f_u(T) \in R^\times$ such that

$$f_u(\zeta_n - 1) = u_n$$

for all n . This is Coleman power series.

2. **Proposition 2.2.3.** There are unique continuous maps $N : R \rightarrow R$ and $\psi : R \rightarrow R$ such that

$$(\varphi \circ N)(f)(T) = \prod_{\xi \in \mu_p} f(\xi(1+T) - 1), \quad (2.1)$$

and

$$(\varphi \circ \psi)(f)(T) = \frac{1}{p} \sum_{\xi \in \mu_p} f(\xi(1+T) - 1). \quad (2.2)$$

Moreover ψ is $\mathbb{Z}_p$ -linear, $\psi \circ \varphi = 1_R$, and N preserves products.

3. **Definition 2.3.6 and Corollary 2.3.7.** Define

$$W = \{f \in R^\times : N(f) = f\}.$$

Then

$$U_\infty \xrightarrow{\sim} W, \quad u \mapsto f_u(T)$$

is a G -isomorphism.

4. **Definition 2.4.4 and Theorem 2.4.6.** For $f \in R^\times$,

$$\Delta(f) = \frac{(1+T)f'(T)}{f(T)}.$$

The theorem states

$$\Delta(W) = R^{\psi=1}.$$

This is the delicate input in the proof of Theorem 2.5.2.

5. **Lemma 2.5.1.** For $f \in R^\times$, define

$$L(f) = \frac{1}{p} \log \left(\frac{f(T)^p}{\varphi(f)(T)} \right). \quad (2.10)$$

Then $L(f) \in R$, and if $f \in W$, then $L(f) \in R^{\psi=0}$. Thus $L : W \rightarrow R^{\psi=0}$ is a G -homomorphism.

6. **Theorem 2.5.2.** Let

$$A = \{\xi(1+T)^a : \xi \in \mu_{p-1}, a \in \mathbb{Z}_p\}.$$

There is a canonical exact sequence of G -modules

$$0 \longrightarrow A \longrightarrow W \xrightarrow{L} R^{\psi=0} \xrightarrow{\alpha} \mathbb{Z}_p \longrightarrow 0, \quad (2.11)$$

where

$$\alpha(f) = (Df)(0).$$

7. **Definition 2.6.1.** For $k \geq 1$,

$$\delta_k(u) = \left(D^{k-1} \frac{(1+T)f'_u(T)}{f_u(T)} \right)_{T=0}. \quad (2.12)$$

8. **Lemma 2.6.2 and identity (2.13).** The map δ_k is a homomorphism and satisfies

$$\delta_k(\sigma u) = \chi(\sigma)^k \delta_k(u).$$

Also, for $a \in \mathbb{Z}_p$,

$$D^k(g((1+T)^a - 1)) = a^k (D^k g)((1+T)^a - 1). \quad (2.13)$$

9. **Theorem 2.6.4.** For $k = 1, \dots, p-1$,

$$\delta_k(U_\infty) = \mathbb{Z}_p.$$

This is used in Section 3.6 for the case $k = 1$.

10. **Section 3.2.** Elements of $\Lambda(G)$ are bounded $\mathbb{Z}_p$ -valued measures on G . Conversely, a linear functional $L : C(G, \mathbb{C}_p) \rightarrow \mathbb{C}_p$ satisfying

$$|L(f)|_p \leq \|f\|$$

for all continuous f , and satisfying $L(f) \in \mathbb{Q}_p$ when f is $\mathbb{Q}_p$ -valued, is uniquely of the form $L = M_\lambda$ for some $\lambda \in \Lambda(G)$. This converse is important in Section 3.4: whenever we define a new bounded functional by a formula such as

$$f \mapsto \int_{\mathbb{Z}_p} f \mathbf{1}_X d\lambda,$$

Section 3.2 guarantees that this functional is represented by a unique measure in $\Lambda(\mathbb{Z}_p)$.

11. **Theorem 3.3.1, Mahler's theorem.** Every continuous $f : \mathbb{Z}_p \rightarrow \mathbb{C}_p$ has a unique expansion

$$f(x) = \sum_{n \geq 0} a_n \binom{x}{n}, \quad a_n \rightarrow 0. \quad (3.5)$$

12. **Definition 3.3.2 and Theorem 3.3.3.** For $\lambda \in \Lambda(\mathbb{Z}_p)$,

$$M(\lambda) = \sum_{n \geq 0} \left(\int_{\mathbb{Z}_p} \binom{x}{n} d\lambda \right) T^n.$$

The Mahler transform

$$M : \Lambda(\mathbb{Z}_p) \xrightarrow{\sim} R$$

is an isomorphism of $\mathbb{Z}_p$ -algebras.

13. **Lemma 3.3.5.** If $g \in R$ and $\Upsilon(g) \in \Lambda(\mathbb{Z}_p)$ is its inverse Mahler transform, then for all $k \geq 0$,

$$\int_{\mathbb{Z}_p} x^k d\Upsilon(g) = (D^k g)(0).$$

This is the essential computational tool in Proposition 3.5.2.

3 Section 3.4: Restriction of Measures

The point of Section 3.4 is that although $\mathbb{Z}_p^\times$ is not a subgroup of the additive group $\mathbb{Z}_p$, there is still a canonical way to regard measures on $\mathbb{Z}_p^\times$ as measures on $\mathbb{Z}_p$ supported on $\mathbb{Z}_p^\times$.

Before proving Lemmas 3.4.1 and 3.4.2, it is useful to record explicitly how standard operations on measures are reflected under the Mahler transform. Recall that

$$M(\lambda)(T) = \int_{\mathbb{Z}_p} (1+T)^x d\lambda(x), \quad (1+T)^x = \sum_{n \geq 0} \binom{x}{n} T^n.$$

3.1 Multiplication by x

Let $\lambda \in \Lambda(\mathbb{Z}_p)$. Define the measure $x\lambda$ by

$$\int_{\mathbb{Z}_p} f(x) d(x\lambda)(x) = \int_{\mathbb{Z}_p} xf(x) d\lambda(x).$$

Then

$$M(x\lambda) = DM(\lambda), \quad D = (1+T) \frac{d}{dT}.$$

Indeed,

$$M(x\lambda)(T) = \int_{\mathbb{Z}_p} x(1+T)^x d\lambda(x),$$

whereas

$$DM(\lambda)(T) = (1+T) \frac{d}{dT} \int_{\mathbb{Z}_p} (1+T)^x d\lambda(x) = \int_{\mathbb{Z}_p} x(1+T)^x d\lambda(x).$$

Equivalently, on Mahler coefficients this is the identity

$$x \binom{x}{n} = (x-n) \binom{x}{n} + n \binom{x}{n} = (n+1) \binom{x}{n+1} + n \binom{x}{n}.$$

Corollary 1. For $\lambda \in \Lambda(\mathbb{Z}_p)$ and $k \geq 0$,

$$\int_{\mathbb{Z}_p} x^k d\lambda = (D^k M(\lambda))(0).$$

3.2 Multiplication by z^x

More generally, if $g \in C(\mathbb{Z}_p, \mathbb{C}_p)$ and λ is a measure, define $g\lambda$ by

$$\int_{\mathbb{Z}_p} f d(g\lambda) = \int_{\mathbb{Z}_p} fg d\lambda.$$

The special case $g(x) = z^x$ is useful. If $z \in \mathbb{C}_p$ satisfies $|z-1|_p < 1$, then

$$z^x = \sum_{n \geq 0} \binom{x}{n} (z-1)^n$$

is a continuous function of $x \in \mathbb{Z}_p$. Then

$$M(z^x \lambda)(T) = M(\lambda)((1+T)z-1).$$

Indeed,

$$M(z^x \lambda)(T) = \int_{\mathbb{Z}_p} z^x (1+T)^x d\lambda(x) = \int_{\mathbb{Z}_p} ((1+T)z)^x d\lambda(x) = M(\lambda)((1+T)z-1).$$

3.3 Restriction to Open Compact Subsets

Let $X \subset \mathbb{Z}_p$ be open and compact, and let $\mathbf{1}_X$ be its characteristic function. Define the restriction of λ to X by

$$\int_{\mathbb{Z}_p} f d\text{Res}_X(\lambda) = \int_{\mathbb{Z}_p} f \mathbf{1}_X d\lambda.$$

This definition is legitimate by the converse measure statement in Section 3.2: the right hand side is a bounded functional on $C(\mathbb{Z}_p, \mathbb{C}_p)$ and takes $\mathbb{Q}_p$ -values on $\mathbb{Q}_p$ -valued functions, hence comes from a unique element of $\Lambda(\mathbb{Z}_p)$.

Now take $X = b + p^n \mathbb{Z}_p$. After extending scalars to a finite extension containing μ_{p^n} , the characteristic function is

$$\mathbf{1}_{b+p^n \mathbb{Z}_p}(x) = \frac{1}{p^n} \sum_{\xi \in \mu_{p^n}} \xi^{x-b}.$$

¹ Therefore

$$M(\text{Res}_{b+p^n \mathbb{Z}_p}(\lambda))(T) = \frac{1}{p^n} \sum_{\xi \in \mu_{p^n}} \xi^{-b} M(\lambda)((1+T)\xi - 1). \quad (3.4.A)$$

This is the explicit restriction formula.

3.4 Restriction to $\mathbb{Z}_p^\times$

Since

$$\mathbb{Z}_p^\times = \mathbb{Z}_p \setminus p\mathbb{Z}_p,$$

we get, from the previous formula with $b = 0$ and $n = 1$,

$$M(\text{Res}_{\mathbb{Z}_p^\times}(\lambda))(T) = M(\lambda)(T) - \frac{1}{p} \sum_{\xi \in \mu_p} M(\lambda)((1+T)\xi - 1). \quad (3.4.B)$$

Define the operator $S : R \longrightarrow R$ by

$$S(F)(T) = F(T) - \frac{1}{p} \sum_{\xi \in \mu_p} F(\xi(1+T) - 1) = 1 - \varphi \circ \psi.$$

Thus

$$M(\text{Res}_{\mathbb{Z}_p^\times}(\lambda)) = S(M(\lambda)).$$

3.5 The Operators φ and ψ

For $a \in \mathbb{Z}_p$, define the pushforward measure $\sigma_a(\lambda)$ by

$$\int_{\mathbb{Z}_p} f(x) d\sigma_a(\lambda)(x) = \int_{\mathbb{Z}_p} f(ax) d\lambda(x).$$

Then

$$M(\sigma_a(\lambda))(T) = M(\lambda)((1+T)^a - 1).$$

This is the same formula as the G -action on R via the cyclotomic character.

Similarly define the measure $\varphi(\lambda)$ by pushforward under $x \mapsto px$:

$$\int_{\mathbb{Z}_p} f(x) d\varphi(\lambda)(x) = \int_{\mathbb{Z}_p} f(px) d\lambda(x).$$

¹This is the finite Fourier orthogonality relation on $\mathbb{Z}/p^n\mathbb{Z}$. If $x - b \equiv 0 \pmod{p^n}$, then every term is 1 and the average is 1. If $x - b \not\equiv 0 \pmod{p^n}$, choose a primitive p^n -th root ζ ; then $\sum_{j=0}^{p^n-1} (\zeta^{x-b})^j = 0$, so the average is 0.

Then

$$M(\varphi(\lambda))(T) = M(\lambda)((1+T)^p - 1) = \varphi(M(\lambda))(T).$$

Define $\psi(\lambda)$ by

$$\int_{\mathbb{Z}_p} f(x) d\psi(\lambda)(x) = \int_{p\mathbb{Z}_p} f(p^{-1}x) d\lambda(x).$$

Then

$$\psi \circ \varphi = \text{id}, \quad \varphi \circ \psi(\lambda) = \text{Res}_{p\mathbb{Z}_p}(\lambda).$$

Indeed,

$$\begin{aligned} \int_{\mathbb{Z}_p} f(x) d(\psi \circ \varphi)(\lambda) &= \int_{p\mathbb{Z}_p} f(p^{-1}x) d\varphi(\lambda)(x) \\ &= \int_{\mathbb{Z}_p} f(p^{-1}px) d\lambda(x) = \int_{\mathbb{Z}_p} f(x) d\lambda(x), \end{aligned}$$

and

$$\begin{aligned} \int_{\mathbb{Z}_p} f(x) d(\varphi \circ \psi)(\lambda) &= \int_{\mathbb{Z}_p} f(px) d\psi(\lambda)(x) \\ &= \int_{p\mathbb{Z}_p} f(x) d\lambda(x) = \int_{\mathbb{Z}_p} f(x) d\text{Res}_{p\mathbb{Z}_p}(\lambda)(x). \end{aligned}$$

Consequently,

$$\text{Res}_{\mathbb{Z}_p^\times}(\lambda) = (1 - \varphi \circ \psi)(\lambda).$$

Under Mahler transform,

$$M((\varphi \circ \psi)(\lambda))(T) = \frac{1}{p} \sum_{\xi \in \mu_p} M(\lambda)((1+T)\xi - 1).$$

Equivalently,

$$M((\varphi \circ \psi)(\lambda))(T) = \varphi(\psi(M(\lambda)))(T).$$

Here the notation φ, ψ is being used on both sides: on the measure side they are defined by pushforward/restriction as above, and on the power series side they are the operators from the trace formula. Since $M(\varphi(\lambda)) = \varphi(M(\lambda))$ and φ is injective on R , this identity implies

$$M(\psi(\lambda)) = \psi(M(\lambda)).$$

Corollary 2. *A measure $\lambda \in \Lambda(\mathbb{Z}_p)$ is supported on $\mathbb{Z}_p^\times$ if and only if*

$$\psi(M(\lambda)) = 0.$$

Proof. The measure λ is supported on $\mathbb{Z}_p^\times$ if and only if $\text{Res}_{\mathbb{Z}_p^\times}(\lambda) = \lambda$, equivalently

$$(1 - \varphi\psi)(\lambda) = \lambda.$$

This is equivalent to $\varphi\psi(\lambda) = 0$. Since φ is injective on measures, this is equivalent to $\psi(\lambda) = 0$. Applying the Mahler transform and using $M(\psi(\lambda)) = \psi(M(\lambda))$ gives $\psi(M(\lambda)) = 0$. $\square$

Lemma 1 (Coates–Sujatha, Lemma 3.4.1). *For every $\lambda \in \Lambda(\mathbb{Z}_p)$,*

$$S(M(\lambda)) = M(\text{Res}_{\mathbb{Z}_p^\times}(\lambda)).$$

In particular,

$$\text{Res}_{\mathbb{Z}_p^\times}(\lambda) = \lambda \iff M(\lambda) \in R^{\psi=0}.$$

Finite-level proof. At finite level, write the image of λ in $\mathbb{Z}_p[\mathbb{Z}_p/p^n\mathbb{Z}_p]$ as

$$\mathrm{pr}_n(\lambda) = \sum_{k=0}^{p^n-1} e_n(k)(k + p^n\mathbb{Z}_p). \quad (3.9)$$

The coefficient $e_n(k)$ is the measure of the residue class $k + p^n\mathbb{Z}_p$:

$$e_n(k) = \int_{\mathbb{Z}_p} \mathbf{1}_{k+p^n\mathbb{Z}_p} d\lambda.$$

Applying $\mathrm{Res}_{\mathbb{Z}_p^\times}$ simply deletes the terms with $p \mid k$:

$$\mathrm{pr}_n(\mathrm{Res}_{\mathbb{Z}_p^\times}(\lambda)) = \sum_{\substack{0 \leq k \leq p^n-1 \\ (k,p)=1}} e_n(k)(k + p^n\mathbb{Z}_p).$$

Let

$$\omega_n(T) = (1 + T)^{p^n} - 1.$$

The finite-level Mahler transform is the algebra map

$$M_n : \mathbb{Z}_p[\mathbb{Z}_p/p^n\mathbb{Z}_p] \longrightarrow R/\omega_n R$$

defined on group-like basis elements by

$$M_n(k + p^n\mathbb{Z}_p) = (1 + T)^k \pmod{\omega_n}.$$

It is an isomorphism: if γ_n denote the generator of $\mathbb{Z}_p/p^n\mathbb{Z}_p$, then

$$\mathbb{Z}_p[\mathbb{Z}_p/p^n\mathbb{Z}_p] \cong \mathbb{Z}_p[X]/(X^{p^n} - 1), \quad k + p^n\mathbb{Z}_p \mapsto X^k,$$

also maps γ_n to one p^n -root. Putting $X = 1 + T$, since ω_n is distinguished, we have

$$\mathbb{Z}_p[X]/(X^{p^n} - 1) \cong \mathbb{Z}_p[T]/((1 + T)^{p^n} - 1) = \mathbb{Z}_p[T]/(\omega_n) \cong R/nR.$$

Thus

$$\mathrm{pr}'_n(M(\lambda)) = M_n(\mathrm{pr}_n(\lambda)) \equiv \sum_{k=0}^{p^n-1} e_n(k)(1 + T)^k \pmod{\omega_n}.$$

Similarly,

$$\mathrm{pr}'_n(M(\mathrm{Res}_{\mathbb{Z}_p^\times}(\lambda))) \equiv \sum_{\substack{0 \leq k \leq p^n-1 \\ (k,p)=1}} e_n(k)(1 + T)^k \pmod{\omega_n}.$$

At this finite level the operator $S = 1 - \varphi\psi$ removes exactly the classes divisible by p , so

$$S(M(\lambda)) \equiv M(\mathrm{Res}_{\mathbb{Z}_p^\times}(\lambda)) \pmod{\omega_n}.$$

Since this holds for every n , it holds in the inverse limit R . □

Remark 1 (Relation with the explicit restriction calculation). The calculation preceding the lemma already proves the same identity in a more conceptual way:

$$M(\mathrm{Res}_{\mathbb{Z}_p^\times}(\lambda)) = S(M(\lambda)).$$

The explicit calculation explains the operator S as $1 - \varphi\psi$, i.e. as “remove the part supported on $p\mathbb{Z}_p$ ”. The finite-level proof explains how this is seen inside the Iwasawa algebra itself: at level p^n , a measure is a linear combination of residue classes $k + p^n\mathbb{Z}_p$, and restriction to $\mathbb{Z}_p^\times$ literally deletes the classes with $p \mid k$. Thus the two arguments are the same idea in two languages. The first is the operator language on integration; the second is the projective-limit language of group algebras.

Lemma 2 (Coates–Sujatha, Lemma 3.4.2). *Let*

$$i : \Lambda(\mathbb{Z}_p^\times) \longrightarrow \Lambda(\mathbb{Z}_p)$$

be the natural inclusion, defined by

$$\int_{\mathbb{Z}_p} f \, di(\eta) = \int_{\mathbb{Z}_p^\times} f|_{\mathbb{Z}_p^\times} \, d\eta.$$

Then

$$i(\Lambda(\mathbb{Z}_p^\times)) = \{\lambda \in \Lambda(\mathbb{Z}_p) : \text{Res}_{\mathbb{Z}_p^\times}(\lambda) = \lambda\}.$$

Consequently,

$$M(i(\Lambda(\mathbb{Z}_p^\times))) = R^{\psi=0}.$$

4 Section 3.5: The Fundamental Exact Sequence

Recall that the cyclotomic character is defined by

$$\sigma(\zeta) = \zeta^{\chi(\sigma)}$$

for every $\sigma \in G$ and every p -power root of unity ζ . It gives a canonical isomorphism

$$\chi : G \xrightarrow{\sim} \mathbb{Z}_p^\times.$$

4.1 The G -action on $\Lambda(\mathbb{Z}_p)$ and on R

The group G acts on the additive group $\mathbb{Z}_p$ through χ :

$$a \longmapsto \chi(g)a.$$

If $a_{\mathbb{Z}_p}$ denotes the element of $\Lambda(\mathbb{Z}_p)$ corresponding to $a \in \mathbb{Z}_p$, define

$$g \cdot a_{\mathbb{Z}_p} = (\chi(g)a)_{\mathbb{Z}_p}.$$

By $\mathbb{Z}_p$ -linearity and continuity this extends to all of $\Lambda(\mathbb{Z}_p)$. In finite level notation, if

$$\lambda_n = \sum_{a \bmod p^n} c_a(a + p^n \mathbb{Z}_p),$$

then

$$g \cdot \lambda_n = \sum_{a \bmod p^n} c_a(\chi(g)a + p^n \mathbb{Z}_p).$$

Since $\chi(g) \in \mathbb{Z}_p^\times$, multiplication by $\chi(g)$ preserves $\mathbb{Z}_p^\times$. Therefore measures supported on $\mathbb{Z}_p^\times$ are sent to measures supported on $\mathbb{Z}_p^\times$, and $\Lambda(\mathbb{Z}_p^\times)$ is a G -submodule of $\Lambda(\mathbb{Z}_p)$.

On $R = \mathbb{Z}_p[[T]]$, the G -action is the one from Chapter 2:

$$(gF)(T) = F((1+T)^{\chi(g)} - 1). \tag{2.7}$$

The Mahler transform is compatible with these actions. Indeed,

$$M(1_{\mathbb{Z}_p}) = 1 + T,$$

so, by continuity, $M(a_{\mathbb{Z}_p}) = (1+T)^a$ for $a \in \mathbb{Z}_p$. Hence

$$M(g \cdot a_{\mathbb{Z}_p}) = M((\chi(g)a)_{\mathbb{Z}_p}) = (1+T)^{\chi(g)a},$$

whereas

$$g \cdot M(a_{\mathbb{Z}_p}) = (1 + ((1+T)^{\chi(g)} - 1))^a = (1+T)^{\chi(g)a}.$$

Thus $M(g \cdot \lambda) = g \cdot M(\lambda)$ for all $\lambda \in \Lambda(\mathbb{Z}_p)$, by linearity and continuity. Hence

$$M : \Lambda(\mathbb{Z}_p) \xrightarrow{\sim} R$$

is an isomorphism of G -modules.

4.2 Definition of $\tilde{L}$

It induces an isomorphism of Iwasawa algebras

$$\tilde{\chi} : \Lambda(G) \xrightarrow{\sim} \Lambda(\mathbb{Z}_p^\times). \quad (3.10)$$

Combining this with the inclusion $\Lambda(\mathbb{Z}_p^\times) \subset \Lambda(\mathbb{Z}_p)$ and the Mahler transform gives

$$\tilde{M} := M \circ \tilde{\chi} : \Lambda(G) \xrightarrow{\sim} R^{\psi=0}.$$

For $u = (u_n) \in U_\infty$, recall that the Coleman power series f_u is the unique element of $R^\times$ satisfying

$$f_u(\zeta_n - 1) = u_n \quad (n \geq 0).$$

Now define

$$\tilde{L}(u) = \tilde{M}^{-1}(L(f_u)).$$

Equivalently,

$$L(f_u) = \sum_{n \geq 0} T^n \int_G \binom{\chi(g)}{n} d\tilde{L}(u).$$

Theorem 1 (Coates–Sujatha, Theorem 3.5.1). *There is an exact sequence of G -modules*

$$0 \longrightarrow \mu_{p-1} \times T_p(\mu) \longrightarrow U_\infty \xrightarrow{\tilde{L}} \Lambda(G) \xrightarrow{\beta} T_p(\mu) \longrightarrow 0, \quad (3.11)$$

where the left map is the natural inclusion and the map β is given by

$$\beta(\lambda) = (\zeta_n) \int_G \chi^{d\lambda}.$$

That is, if

$$a = \int_G \chi(g) d\lambda \in \mathbb{Z}_p,$$

then

$$\beta(\lambda) = (\zeta_n^a)_n \in T_p(\mu).$$

Proof. The proof is a translation of the exact sequence for Coleman power series:

$$0 \longrightarrow A \longrightarrow W \xrightarrow{L} R^{\psi=0} \xrightarrow{\alpha} \mathbb{Z}_p \longrightarrow 0,$$

where

$$A = \{\xi(1+T)^a : \xi \in \mu_{p-1}, a \in \mathbb{Z}_p\}, \quad \alpha(F) = (DF)(0).$$

First, use the Coleman isomorphism

$$U_\infty \longrightarrow W, \quad u \longmapsto f_u(T)$$

which says that norm-compatible unit systems are exactly the power series fixed by the norm operator N . Second, by Section 3.4 and the cyclotomic character, the map

$$\tilde{M} : \Lambda(G) \xrightarrow{\sim} R^{\psi=0}$$

is a G -isomorphism. By definition of $\tilde{L}$, the diagram

$$\begin{array}{ccc} U_\infty & \xrightarrow{\tilde{L}} & \Lambda(G) \\ w \mapsto f_u \downarrow \sim & & \downarrow \tilde{M} \\ W & \xrightarrow{L} & R^{\psi=0} \end{array}$$

commutes. Equivalently,

$$\widetilde{M}(\widetilde{L}(u)) = L(f_u).$$

Transporting this exact sequence through the two isomorphisms above gives exactness at U_∞ and at $\Lambda(G)$.

It remains to identify the two end terms explicitly.

First, in the exact sequence just recalled, the kernel of L is

$$A = \{\xi(1+T)^a : \xi \in \mu_{p-1}, a \in \mathbb{Z}_p\}.$$

Under the Coleman isomorphism $U_\infty \simeq W$, the factor $\xi \in \mu_{p-1}$ is the constant root-of-unity system. The factor $(1+T)^a$ corresponds to the compatible system $(\zeta_n^a)_n$, because

$$(1 + (\zeta_n - 1))^a = \zeta_n^a.$$

This is exactly $T_p(\mu)$. Therefore

$$\text{Ker}(\widetilde{L}) = \mu_{p-1} \times T_p(\mu).$$

Second, we compare the map α with the integral $\lambda \mapsto \int_G \chi d\lambda$. Let $\lambda \in \Lambda(G)$ and put

$$F = \widetilde{M}(\lambda) \in R^{\psi=0}.$$

By definition of $\widetilde{M}$, the corresponding measure on $\mathbb{Z}_p$ is $\widetilde{\chi}(\lambda)$, viewed as supported on $\mathbb{Z}_p^\times$. We use the Mahler-transform calculation

$$\int_{\mathbb{Z}_p} x^k d\lambda_0 = (D^k M(\lambda_0))(0),$$

applied with $k = 1$. It gives

$$(DF)(0) = \int_{\mathbb{Z}_p} x d\widetilde{\chi}(\lambda).$$

Since $\widetilde{\chi}$ is induced by $\chi : G \simeq \mathbb{Z}_p^\times$, this equals

$$\int_G \chi(g) d\lambda.$$

Thus the map α corresponds exactly to

$$\lambda \longmapsto \int_G \chi d\lambda.$$

Finally, identifying $\mathbb{Z}_p$ with $T_p(\mu)$ by $a \longmapsto (\zeta_n^a)_n$ gives the stated map β . □

Remark 2. The exactness at $\Lambda(G)$ can be stated very concretely:

$$\lambda \in \text{Im}(\widetilde{L}) \iff \int_G \chi(g) d\lambda = 0.$$

5 Proposition 3.5.2: Integrating Characters Against $\widetilde{L}(u)$

Proposition 1 (Coates–Sujatha, Proposition 3.5.2). *For all integers $k \geq 1$ and all $u \in U_\infty$,*

$$\int_G \chi(g)^k d\widetilde{L}(u) = (1 - p^{k-1})\delta_k(u). \tag{3.12}$$

Proof. Let $u \in U_\infty$ and put

$$h_u(T) = \frac{(1+T)f'_u(T)}{f_u(T)} = \Delta(f_u).$$

By definition,

$$\delta_k(u) = \left(D^{k-1} h_u(T) \right)_{T=0}. \quad (1)$$

We first rewrite the left-hand side as an integral over $\mathbb{Z}_p^\times$. For any $\lambda \in \Lambda(G)$, the isomorphism $\tilde{\chi} : \Lambda(G) \rightarrow \Lambda(\mathbb{Z}_p^\times)$ gives

$$\int_G \chi(g)^k d\lambda = \int_{\mathbb{Z}_p^\times} x^k d\tilde{\chi}(\lambda). \quad (2)$$

When $\tilde{\chi}(\lambda)$ is regarded as a measure on $\mathbb{Z}_p$ supported on $\mathbb{Z}_p^\times$, the same integral is

$$\int_{\mathbb{Z}_p} x^k d\tilde{\chi}(\lambda). \quad (3)$$

Now take $\lambda = \tilde{L}(u)$. By definition of $\tilde{L}$,

$$\tilde{M}(\tilde{L}(u)) = L(f_u).$$

Since $\tilde{M} = M \circ \tilde{\chi}$, this means

$$M(\tilde{\chi}(\tilde{L}(u))) = L(f_u),$$

or equivalently

$$\tilde{\chi}(\tilde{L}(u)) = \Upsilon(L(f_u)).$$

Therefore the Mahler-transform formula (recall that $M = \Upsilon^{-1}$)

$$\int_{\mathbb{Z}_p} x^k d\Upsilon(g) = (D^k g)(0)$$

gives

$$\int_G \chi(g)^k d\tilde{L}(u) = \int_{\mathbb{Z}_p} x^k d\Upsilon(L(f_u)) = (D^k L(f_u))(0). \quad (4)$$

It remains to compute $D^k L(f_u)$. By definition,

$$L(f_u) = \frac{1}{p} \log \left(\frac{f_u(T)^p}{\varphi(f_u)(T)} \right).$$

Apply $D = (1+T)d/dT$. Since $D \log(f) = Df/f$ and $Df = (1+T)f'$, we get

$$DL(f_u) = \frac{1}{p} \left(p \frac{Df_u}{f_u} - \frac{D\varphi(f_u)}{\varphi(f_u)} \right).$$

The first term is $h_u(T)$. For the second term, observe that

$$\varphi(f_u)(T) = f_u((1+T)^p - 1).$$

Using the chain rule,

$$\frac{D\varphi(f_u)}{\varphi(f_u)} = p \varphi \left(\frac{Df_u}{f_u} \right) = p \varphi(h_u).$$

Hence

$$DL(f_u) = h_u - \varphi(h_u). \quad (5)$$

Applying D^{k-1} to both sides and evaluating at $T = 0$ gives

$$(D^k L(f_u))(0) = (D^{k-1} h_u)(0) - (D^{k-1} \varphi(h_u))(0). \quad (6)$$

Here we use the elementary identity

$$D^m(g((1+T)^a - 1)) = a^m(D^m g)((1+T)^a - 1), \quad (2.13)$$

which follows by induction from the chain rule and from the definition $D = (1+T)d/dT$. Applying this with $m = k-1$, $a = p$, and $g = h_u$, we get

$$D^{k-1}(h_u((1+T)^p - 1)) = p^{k-1}(D^{k-1}h_u)((1+T)^p - 1).$$

Evaluating at $T = 0$ gives

$$(D^{k-1}\varphi(h_u))(0) = p^{k-1}(D^{k-1}h_u)(0). \quad (7)$$

Combining (6), (7), and (1), we obtain

$$(D^k L(f_u))(0) = (1 - p^{k-1})(D^{k-1}h_u)(0) = (1 - p^{k-1})\delta_k(u).$$

Together with (4), this proves the proposition. $\square$

6 Section 3.6: The Image of δ_k

Section 3.6 computes $\delta_k(U_\infty)$ for every $k \geq 1$. The new ingredient is a decomposition of $\Lambda(G)$ into $p-1$ copies of R .

The decomposition comes from the standard splitting

$$\mathbb{Z}_p^\times \simeq \mu_{p-1} \times (1 + p\mathbb{Z}_p).$$

Pull this back through the cyclotomic character. Write

$$G = \langle \varpi \rangle \times \Gamma, \quad (3.13)$$

where

$$\langle \varpi \rangle = \chi^{-1}(\mu_{p-1}) \text{ is cyclic of order } p-1, \quad \Gamma = \chi^{-1}(1 + p\mathbb{Z}_p) \simeq \mathbb{Z}_p.$$

Fix a topological generator γ of Γ . Let

$$A = \mathbb{Z}_p[\langle \varpi \rangle].$$

Then

$$\Lambda(G) \simeq A[[T]],$$

where γ corresponds to $1+T$. Since the characters of $\langle \varpi \rangle$ are the powers of

$$\chi_\varpi := \chi|_{\langle \varpi \rangle},$$

one gets

$$A[[T]] \simeq \bigoplus_{i \bmod p-1} R.$$

Explicitly, if

$$f(T) = \sum_{n \geq 0} a_n T^n \in A[[T]], \quad a_n \in A,$$

then its image is the tuple

$$\left(\sum_{n \geq 0} \chi_\varpi^i(a_n) T^n \right)_{i \bmod p-1}.$$

Equivalently, this is the decomposition of the group ring $A = \mathbb{Z}_p[\langle \varpi \rangle]$ into its character components with respect to the characters χ_ϖ^i . Let

$$\vartheta : \Lambda(G) \xrightarrow{\sim} \bigoplus_{i \bmod p-1} R$$

be this isomorphism, and write $\vartheta(\lambda)_i$ for the i -th component.

Lemma 3 (Coates–Sujatha, Lemma 3.6.2). *For all $\lambda \in \Lambda(G)$ and all integers $k \geq 1$,*

$$\int_G \chi(g)^k d\lambda = \vartheta(\lambda)_i (\chi(\gamma)^k - 1),$$

where

$$i \equiv k \pmod{p-1}.$$

Proof. Write λ in $A[[T]]$ as

$$\lambda = \sum_{n \geq 0} a_n T^n, \quad a_n \in \mathbb{Z}_p[\varpi].$$

The character χ^k evaluates a_n by applying χ_ϖ^k on $\langle \varpi \rangle$, and evaluates T by

$$T \mapsto \chi(\gamma)^k - 1.$$

Thus

$$\chi^k(\lambda) = \sum_{n \geq 0} \chi^k(a_n) (\chi(\gamma)^k - 1)^n.$$

Since χ_ϖ^k depends only on $i \equiv k \pmod{p-1}$, this is exactly

$$\vartheta(\lambda)_i (\chi(\gamma)^k - 1).$$

But by the measure interpretation,

$$\chi^k(\lambda) = \int_G \chi(g)^k d\lambda.$$

□

Corollary 3 (Coates–Sujatha, Corollary 3.6.3). *The image of $\tilde{L}$ under ϑ is the subset of $\bigoplus_{i \bmod p-1} R$ consisting of tuples (α_i) such that*

$$\alpha_i \in R \quad \text{if } i \not\equiv 1 \pmod{p-1},$$

and

$$\alpha_1 \in ((1 + T) - \chi(\gamma))R.$$

Proof. By the fundamental exact sequence,

$$\text{Im}(\tilde{L}) = \text{Ker}(\beta).$$

The map β is determined by the integral against χ

$$\lambda \mapsto \int_G \chi(g) d\lambda.$$

By the componentwise integral formula with $k = 1$,

$$\int_G \chi(g) d\lambda = \vartheta(\lambda)_1 (\chi(\gamma) - 1).$$

Thus λ lies in $\text{Ker}(\beta)$ precisely when $\vartheta(\lambda)_1$ vanishes at

$$T = \chi(\gamma) - 1.$$

Equivalently,

$$\vartheta(\lambda)_1 \in (T - (\chi(\gamma) - 1))R = ((1 + T) - \chi(\gamma))R.$$

The other components are unrestricted.

□

Before proving the full image statement, recall the earlier result from Chapter 2:

$$\delta_k(U_\infty) = \mathbb{Z}_p \quad (k = 1, \dots, p-1). \quad (2.6.4)$$

This will handle the exceptional case $k = 1$ below.

Proposition 2 (Coates–Sujatha, Proposition 3.6.1). *Let $k \geq 1$. If $k = 1$ or*

$$k \not\equiv 1 \pmod{p-1},$$

then

$$\delta_k(U_\infty) = \mathbb{Z}_p.$$

If $k > 1$ and

$$k \equiv 1 \pmod{p-1},$$

then

$$\delta_k(U_\infty) = p^m \mathbb{Z}_p, \quad m = 1 + \text{ord}_p(k-1).$$

Proof. The case $k = 1$ is already covered by the earlier result saying that $\delta_k(U_\infty) = \mathbb{Z}_p$ for $1 \leq k \leq p-1$, so assume $k > 1$. Then $1 - p^{k-1}$ is a p -adic unit. Hence the formula

$$\int_G \chi(g)^k d\tilde{L}(u) = (1 - p^{k-1})\delta_k(u)$$

shows that the image of δ_k is the same ideal of $\mathbb{Z}_p$ as the set of integrals

$$\left\{ \int_G \chi(g)^k d\lambda : \lambda \in \text{Im}(\tilde{L}) \right\}. \quad (8)$$

Let $i \equiv k \pmod{p-1}$. By the componentwise integral formula,

$$\int_G \chi(g)^k d\lambda = \vartheta(\lambda)_i (\chi(\gamma)^k - 1). \quad (9)$$

Now use Corollary 3.6.3.

If $i \not\equiv 1$, then the i -th component of $\vartheta(\text{Im } \tilde{L})$ is all of R . Hence the possible values in (9) are

$$\{h(\chi(\gamma)^k - 1) : h \in R\}.$$

Since $\chi(\gamma)^k - 1 \in p\mathbb{Z}_p$, evaluation at this point maps $R = \mathbb{Z}_p[[T]]$ onto $\mathbb{Z}_p$ by taking constant terms. Therefore

$$\delta_k(U_\infty) = \mathbb{Z}_p.$$

Now suppose $i \equiv 1$, i.e. $k \equiv 1 \pmod{p-1}$. By Corollary 3.6.3, the first component of an element in $\vartheta(\text{Im } \tilde{L})$ has the form

$$((1+T) - \chi(\gamma))h(T), \quad h(T) \in R.$$

Evaluating at $T = \chi(\gamma)^k - 1$ gives

$$(\chi(\gamma)^k - \chi(\gamma))h(\chi(\gamma)^k - 1).$$

Since $h(\chi(\gamma)^k - 1)$ runs through all of $\mathbb{Z}_p$, the image is

$$(\chi(\gamma)^k - \chi(\gamma))\mathbb{Z}_p.$$

Write

$$\chi(\gamma) = 1 + pw, \quad w \in \mathbb{Z}_p^\times.$$

Then

$$\chi(\gamma)^k - \chi(\gamma) = \chi(\gamma)(\chi(\gamma)^{k-1} - 1).$$

Since $\chi(\gamma)$ is a unit, the valuation is

$$\text{ord}_p(\chi(\gamma)^k - \chi(\gamma)) = \text{ord}_p((1 + pw)^{k-1} - 1).$$

For $k > 1$ this valuation is

$$1 + \text{ord}_p(k - 1),$$

by the standard binomial valuation formula for $1 + pw$ with $w \in \mathbb{Z}_p^\times$. Hence

$$\delta_k(U_\infty) = p^{1+\text{ord}_p(k-1)}\mathbb{Z}_p.$$

□